

Supplementary Material: Extended Experimental Results, Statistical Analysis, and Data Integrity Verification

Table of Contents

Part A: Supplementary by Subsection and Table Number	1
Section 1: Extended Performance Evaluation and Ablation Studies.....	1
Section 2: Statistical Significance and Per-Class Analysis.....	2
Section 3: Controlled Environment and Edge Case Analysis.....	4
Part B: Raw Data Repository Index.....	5

Part A: Extended Tables

Section 1: Extended Performance Evaluation and Ablation Studies

1.1: Extended Attention Ablation Study Metrics

Table S1: Detailed ablation performance across three random seeds. Metrics report number of parameters, overall accuracy, and weighted F1-score formatted to decimal places (mean $\pm$ std).

model	name	seed	params	acc	f1
none	CNN (base)	0	167811	0.8971036585365854	0.896414046375126
none	CNN (base)	1	167811	0.9115853658536586	0.91124662017805
none	CNN (base)	2	167811	0.9233993902439024	0.9239188370390325
channel	CNN + Channel	0	171011	0.9115853658536586	0.9104794615713756
channel	CNN + Channel	1	171011	0.9264481707317073	0.9261308511444478
channel	CNN + Channel	2	171011	0.9249237804878049	0.9243776361613253
spatial	CNN + Spatial	0	168203	0.9085365853658537	0.9093032176465787
spatial	CNN + Spatial	1	168203	0.911204268292683	0.9112786655345607
spatial	CNN + Spatial	2	168203	0.9230182926829268	0.9230747729557661
cbam	CNN + CBAM	0	171403	0.9222560975609756	0.9224563444010069
cbam	CNN + CBAM	1	171403	0.9047256097560976	0.9048770674969837
cbam	CNN + CBAM	2	171403	0.9253048780487805	0.9252910897056652
coord	CNN + Coord. Attn.	0	175395	0.9401676829268293	0.9400500430828728
coord	CNN + Coord. Attn.	1	175395	0.9355945121951219	0.9351946730879686
coord	CNN + Coord. Attn.	2	175395	0.9409298780487805	0.94085517558986

1.2: Comprehensive Baseline Comparison

Table S2: *Extended comparison against baseline models. Results include absolute accuracy, F1-scores, total parameter counts, and relative parameter size scaling compared to the proposed architecture.*

model	name	seed	params	acc	f1
vit_tiny	ViT-Tiny	0	5524995	0.9039634146341463	0.903125690496665
vit_tiny	ViT-Tiny	1	5524995	0.9157774390243902	0.9151715774102533
vit_tiny	ViT-Tiny	2	5524995	0.9146341463414634	0.9147370910242285
resnet_ft	ResNet-50	0	23770691	0.9237804878048781	0.9230757022210384
resnet_ft	ResNet-50	1	23770691	0.9184451219512195	0.9176040662849806
resnet_ft	ResNet-50	2	23770691	0.9245426829268293	0.9248579931077847
dopplernet	DopplerNet (sing. fr.)	0	1096387	0.924161585365853	0.924090498823684
dopplernet	DopplerNet (sing. fr.)	1	1096387	0.930640243902439	0.930706568652371
dopplernet	DopplerNet (sing. fr.)	2	1096387	0.931021341463414	0.930840424462536
coord	CNN + Coord. Attn.	0	175395	0.9401676829268293	0.9400500430828728
coord	CNN + Coord. Attn.	1	175395	0.9355945121951219	0.9351946730879686
coord	CNN + Coord. Attn.	2	175395	0.9409298780487805	0.94085517558986

Section 2: Statistical Significance and Per-Class Analysis

2.1: Unrounded Statistical Significance and Pairwise Hypothesis Testing

The n_{ij} columns read as:

- n_{10} : number of samples where "OURS correct, other wrong"
- n_{01} : number of samples where "other correct, OURS wrong"

Table S3: *Exact statistical significance outputs. The table reports unrounded p-values from McNemar's tests and accuracies across evaluated configurations (seed=0).*

comparator	ours2_acc	other_acc	n_{10}	n_{01}	p_value	sig
ViT-Tiny	0.9397865853658537	0.9028201219512195	137	40	5.361707668894095e-13	yes
CNN+Spatial	0.9413109756097561	0.9066310975609756	124	33	6.830699309565678e-13	yes
CNN (base)	0.9413109756097561	0.9100609756097561	129	47	1.0241217531265784e-09	yes
ResNet-50	0.9397865853658537	0.9237804878048781	77	35	0.00010700371395686192	yes
CNN+Channel	0.9397865853658537	0.9253048780487805	81	43	0.0008915134791795963	yes
CNN+CBAM	0.9397865853658537	0.9264481707317073	73	38	0.0012503549718006987	yes
DopplerNet (single-frame)	0.9401676829268293	0.926829268292683	97	62	0.007009861918957277	yes

2.2: Per-Class F1-Score Breakdown

Table S4: Granular class-wise performance breakdown for Cars, Drones, and People. Scores report F1-score across three independent runs.

model	name	seed	Cars	Drones	People
none	CNN (base)	0	0.8638655462184874	0.8257261410788381	0.977689638076351
none	CNN (base)	1	0.860313315926893	0.8698710433763188	0.9860696517412936
none	CNN (base)	2	0.8911729141475212	0.8829389788293898	0.9828973843058351
channel	CNN + Channel	0	0.8764742396027312	0.8827674567584881	0.9605009633911368
channel	CNN + Channel	1	0.8886138613861386	0.8977061376317421	0.979692917285785
channel	CNN + Channel	2	0.8852251109701966	0.8953771289537713	0.9797730636408486
spatial	CNN + Spatial	0	0.87017099430019	0.8761682242990654	0.9678078691875319
spatial	CNN + Spatial	1	0.8761682242990654	0.8659254414650098	0.9755854509217738
spatial	CNN + Spatial	2	0.885997521685254	0.8876611418047882	0.9815461346633416
cbam	CNN + CBAM	0	0.8878107944208611	0.8881948782011243	0.977977977977978
cbam	CNN + CBAM	1	0.8683914510686164	0.8455284552845529	0.9809428284854563
cbam	CNN + CBAM	2	0.8890251379521765	0.8898568761667703	0.9830845771144279
resnet_ft	ResNet-50	0	0.8809523809523809	0.8962962962962963	0.9793307086614174
resnet_ft	ResNet-50	1	0.8668852459016393	0.8856304985337243	0.9851337958374629
resnet_ft	ResNet-50	2	0.890337877889745	0.8850574712643678	0.9844611528822055
vit_tiny	ViT-Tiny	0	0.8666666666666667	0.8676844783715013	0.9610894941634242
vit_tiny	ViT-Tiny	1	0.873222016079159	0.8809523809523809	0.9769041769041769
vit_tiny	ViT-Tiny	2	0.8774271844660194	0.8791484032561052	0.9735396904643036
dopplernet	DopplerNet (sing. fr.)	0	0.977689638076351	0.977689638076351	0.977689638076351
dopplernet	DopplerNet (sing. fr.)	1	0.9860696517412936	0.9860696517412936	0.9860696517412936
dopplernet	DopplerNet (sing. fr.)	2	0.9828973843058351	0.9828973843058351	0.9828973843058351
coord	CNN + Coord. Attn.	0	0.9105392156862745	0.9140722291407223	0.9880597014925373
coord	CNN + Coord. Attn.	1	0.9112063686466626	0.9146417445482866	0.9900497512437811
coord	CNN + Coord. Attn.	2	0.9085328422344997	0.9133956386292835	0.9851042701092354

Section 3: Controlled Environment and Edge Case Analysis

3.1: Evaluation of DopplerNet Under Varied Experimental Conditions

Table S5: *Individual seed behavior for DopplerNet across eight distinct evaluation setups, contrasting clean environments against potential data-leakage conditions.*

leakage	stack_frames	split	seed	acc	f1
True	1	random	0	0.98131910026687	0.9813220095033954
True	1	random	1	0.974837971788029	0.9748424201738992
True	1	random	2	0.9786504003049943	0.978657613234241
True	1	group	0	0.9902016205012248	0.9902161686219506
True	1	group	1	0.9902016205012248	0.9902221531145979
True	1	group	2	0.9907669116261542	0.9907743300882939
True	3	random	0	0.980470993681792	0.9804685351883431
True	3	random	1	0.9821941412981046	0.9822242680327775
True	3	random	2	0.980470993681792	0.9804862338194403
True	3	group	0	0.9914724275156339	0.9914781141507258
True	3	group	1	0.9903354178510517	0.9903497575331519
True	3	group	2	0.9869243888573053	0.98696504619498
False	1	random	0	0.936713686618376	0.9369507326058746
False	1	random	1	0.938238658025162	0.9384309446054783
False	1	random	2	0.9348074723598933	0.934917912984489
False	1	group	0	0.9210796036897848	0.9211238986099619
False	1	group	1	0.9234711308507004	0.9235692609215869
False	1	group	2	0.9132217287324906	0.9131888621416107
False	3	random	0	0.9552238805970149	0.9552779683812016
False	3	random	1	0.9575200918484501	0.9576068368386568
False	3	random	2	0.9494833524684271	0.9494611113970962
False	3	group	0	0.9361483007209063	0.936324551865994
False	3	group	1	0.9577754891864058	0.957854540743392
False	3	group	2	0.9433573635427395	0.9434367910384014

Part B: Raw Data Repository Index

To ensure strict reproducibility and transparency, the raw experimental outputs and exact data partitioning records are provided as standalone files within the supplementary package.

- **File 1: split_assignment_supplementary.csv**
 - **Description:** *Complete data partition ledger. Contains row-by-row mapping of every data instance, including its assigned split (Train/Val/Test), ground-truth class, file stem, recording ID, and source CSV. This ledger explicitly verifies the absence of spatial or temporal data leakage between partitions.*
- **File 2: confusion_matrices.json**
 - **Description:** *Raw evaluation matrices. Contains the comprehensive 3-by-3 confusion matrix arrays for all 8 core experimental conditions referenced in the main text.*